

STAT 231 Final Exam Information

Please read the Midterm 1 Information File to remind yourself of the policies and procedures for our exams. This document *only* covers differences between the midterms and the final. If you have a question about the final, please double-check that this has not already been answered in the context of Midterm 1 or 2.

Format

The exam is 150 minutes (2.5 hours) long. There will be 7 questions worth a total of 80 marks. Not all questions will be worth the same number of marks. The front page of the exam will indicate how many marks each question is worth.

Coverage

With some exceptions (see next paragraph), the final covers the material in Chapters 1-8 of the Course Notes, with the exception of Sections 2.4, 5.4, and most of Section 6.5. It also covers any additional material covered in Tutorials 1-5, the Introductory R Tutorial, R Tutorials 1-4, and Assignments 1-4. **Very important:** It is assumed you are familiar with all assignment materials, even if you did not complete every assignment. If you did not complete an assignment, it is presumed that you have nevertheless reviewed its content.

Exceptions: Various topics that were specifically highlighted in class as being excluded within the course will not be covered on the final. We will not provide a comprehensive list of what is not covered. Any topics mentioned in class that are not covered in the final will have been clearly highlighted as excluded from the course at the time of that class. In addition, the list of material covered is detailed below. If a topic was discussed in class, but is not covered on this list, then you may assume it is not covered on the final.

Approximately half of the exam will focus on material that was not covered on Midterms 1 and 2. (To avoid any ambiguity, by “material that was not covered on Midterms 1 and 2”, we do not simply mean material that did not fall under the coverage of those exams, but also material that fell under the coverage of those exams but was not specifically examined on either Midterm 1 or 2.) Note, however, that due to the cumulative nature of the material in the course, that many questions will integrate ideas from various areas of the course materials. For the avoidance of doubt, you are expected to be familiar with all of the above material.

Material Summary

To aid you in preparing for the exam, here is a list of the ideas and definitions you should know:

Empirical Studies (Sections 1.1-1.2)

- Units, populations and processes.
- Variate and types of variates.
- Response versus explanatory variates.
- Attributes of a population or process.
- Types of studies (sample surveys, observational studies, experimental studies).

Numerical Summaries (Section 1.3)

- Measures of location: sample mean, median, mode.
- Measures of variability: sample variance, sample standard deviation, range, IQR.
- Measures of shape: skewness, kurtosis.
- Sample percentiles and quantiles.
- Lower or first quartile, upper or third quartile.

- Five number summary.
- Numerical summaries for bivariate data: sample correlation.
- Relative risk.

Graphical Summaries (Section 1.3)

- Relative frequency histograms.
- Empirical cumulative distribution function.
- Scatterplots, run charts, and boxplots.

Data Analysis and Statistical Models (Sections 1.4-1.5, 2.1)

- Descriptive statistics.
- Discrete statistical models: $\text{Binomial}(n; \theta)$, $\text{NegativeBinomial}(k; \theta)$, $\text{Poisson}(\theta)$, $\text{Geometric}(\theta)$.
- Continuous statistical models: $\text{Exponential}(\theta)$, $G(\mu, \sigma)$.
- Note with the preceding you should be familiar with when these models are applied, but you are not expected to know their probability (density) functions.

Estimation (Sections 2.2, 2.3, 2.5)

- Estimate of a parameter.
- Likelihood functions for discrete distributions.
- Maximum likelihood estimate.
- Relative likelihood function and log likelihood function.
- Likelihood function for a random sample.
- Likelihood functions for continuous distributions.
- Invariance property of maximum likelihood estimates.

Check the Fit of a Model (Section 2.6)

You should know how to check the fit of a model by:

- Calculating expected frequencies using the assumed model (discrete distributions **only**) and comparing them with observed relative frequencies.
- Examining a graph of the probability density function of the assumed model superimposed on a relative frequency histogram for the observed data (continuous distributions).
- Examining a graph of the cumulative distribution function of the assumed model superimposed on a graph of the empirical cumulative distribution function for the observed data (continuous distributions).
- Examining a Q-Q plot. Note that you will not be asked to construct a Q-Q plot, but you should be familiar with how it is constructed in order to answer questions about what the Q-Q plot tells you about a dataset.

PPDAC (Chapter 3)

You should be familiar with the terminology introduced in this chapter and, in particular, be able to read a study description and answer related questions.

- Types of problems (descriptive, causative, predictive).
- Target population, study population, and sample/sampling protocol.

- Study error, sample error, measurement error.

Estimation (Sections 4.1-4.2)

You should know and/or understand the following:

- Definition of a point estimate of a parameter.
- Definition of a point estimator of a parameter.
- Sampling distribution of an estimator.
- In particular: if we draw a sample of size n from a population Y where $E[Y] = \mu$ and $sd(Y) = \sigma$, what affects the variance of our sampling distribution for $\bar{Y}$?
- **Important:** it is assumed you are familiar with the central limit theorem and associated results from STAT 230. Note that this includes standardizing Gaussian variables. In particular, you should be able to use the fact that if $Y \sim G(\mu, \sigma)$ then $\frac{Y - \mu}{\sigma} \sim G(0, 1)$ to answer questions concerning probabilities associated with Y . You **do not** need to be able to use probability distribution tables for this test, however!

Likelihood Intervals (Section 4.3)

You should be familiar with the following concepts:

- Relative likelihood function.
- $100p\%$ likelihood interval.
- Log relative likelihood function.
- How to obtain likelihood intervals from a graph of the relative likelihood function or the log relative likelihood function.
- Behaviour of likelihood functions, log relative likelihood functions, and likelihood intervals as the sample size increases.
- Guidelines for interpretation of likelihood intervals.
- In particular, you should know how to construct the relative likelihood functions for each of the following: Binomial(n, θ); Geometric(θ); Negative Binomial(k, θ); Poisson(θ); Exponential(θ); $G(\mu, \sigma)$, σ known; $G(\mu, \sigma)$, μ known. You could also be asked to derive the relative likelihood function for a ‘new’ distribution, given its probability function or probability density function.

Confidence Intervals (Section 4.4)

Confidence intervals: like likelihood intervals, but different. Make sure you know all the following:

- Definition of a confidence interval.
- Interpretation of a $100p\%$ confidence interval.
- Behaviour of the width of a confidence interval as the sample size, confidence level and (if applicable) population variance change.
- Definition of a pivotal quantity and the pivotal quantities and approximate pivotal quantities detailed in Tables 4.3 and 4.4 of the Course Notes.
- Use of change of variables to demonstrate a random variable is a pivotal quantity (see example in Tutorial 3).
- How to find a confidence interval given a pivotal quantity or approximate pivotal quantity.
- $100p\%$ confidence interval for the mean of a Gaussian distribution with known standard deviation: $\bar{y} \pm a \frac{\sigma}{\sqrt{n}}$ where $P(Z \leq a) = \frac{1+p}{2}$, $Z \sim G(0, 1)$.

- Approximate 100p% confidence interval for Binomial(n, θ) data: $\hat{\theta} \pm a\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}$ where $\hat{\theta} = \frac{y}{n}$ and $P(Z \leq a) = \frac{1+p}{2}$, $Z \sim G(0, 1)$.
- For a Binomial experiment choose the sample size n such that $n \geq \left\lceil \frac{a(0.5)}{\ell} \right\rceil^2$ where $P(Z \leq a) = \frac{1+p}{2}$, $Z \sim G(0, 1)$ to ensure an approximate 100p% confidence interval for θ has width $\leq 2\ell$.
- Approximate 100p% confidence interval for θ for Poisson(θ) data: $\bar{y} \pm a\sqrt{\frac{\bar{y}}{n}}$ where $P(Z \leq a) = \frac{1+p}{2}$, $Z \sim G(0, 1)$.
- Approximate 100p% confidence interval for θ for Exponential(θ) data: $\bar{y} \pm a\frac{\bar{y}}{\sqrt{n}}$ where $P(Z \leq a) = \frac{1+p}{2}$, $Z \sim G(0, 1)$.

Chi-squared and t Distributions (Section 4.5)

You should be familiar with the properties of both distributions:

- If $W_1, W_2, \dots, W_n$ are independent random variables with $W_i \sim \chi_{k_i}^2$, then $S = \sum_{i=1}^n W_i \sim \chi_{\sum k_i}^2$.
- If $Z \sim G(0, 1)$, then $Z^2 = W \sim \chi_1^2$, and so if $W \sim \chi_1^2$, then $P(W \geq w) = 2[1 - P(Z \leq \sqrt{w})]$ where $Z \sim G(0, 1)$.
- If $Z_1, Z_2, \dots, Z_n \sim G(0, 1)$, then $S = \sum_{i=1}^n Z_i^2 \sim \chi_n^2$.
- If $W \sim \chi_2^2$, then $W \sim \text{Exponential}(2)$ and so, if $W \sim \chi_2^2$, then $P(W \geq w) = e^{-w/2}$.
- If $Z \sim G(0, 1)$ and $U \sim \chi_k^2$ independently, then $T = \frac{Z}{\sqrt{U/k}} \sim t_k$.
- If T_k follows a t distribution with k degrees of freedom, as k increases the p.d.f. of T_k approaches that of a $G(0, 1)$ random variable.

Likelihood-Based Confidence Intervals (Section 4.6)

This section covered the likelihood ratio statistic, and how it allows us to relate likelihood and confidence intervals.

- Likelihood ratio statistic and its approximate distribution for large n .
- Likelihood-based confidence intervals, such as a 15% likelihood interval is an approximate 95% confidence interval and a 10% likelihood interval is an approximate 97% confidence interval.

Convert a 100q% confidence interval to a 100p% likelihood interval:

1. Find c such that $q = P(W \leq c) = P(|Z| \leq \sqrt{c}) = 2P(Z \leq \sqrt{c}) - 1$
2. Then $p = e^{-c/2}$

Convert a 100p% likelihood interval to a 100q% confidence interval:

1. $q = 2P(Z \leq \sqrt{-2\log p}) - 1$

Confidence Intervals for Parameters in the $G(\mu, \sigma)$ Model (Section 4.7)

Be familiar with the following:

- 100p% confidence interval for the mean μ of a Gaussian distribution with unknown standard deviation σ : $\bar{y} \pm a\frac{s}{\sqrt{n}}$ where $P(T \leq a) = \frac{1+p}{2}$, $T \sim t_{n-1}$ and s is the sample standard deviation.
- 100p% confidence interval for the variance σ^2 of a Gaussian distribution with unknown mean μ : $\left[\frac{(n-1)s^2}{b}, \frac{(n-1)s^2}{a} \right]$ where $P(W \leq a) = \frac{1-p}{2}$, $P(W \leq b) = \frac{1+p}{2}$, $W \sim \chi_{n-1}^2$.

- For a Gaussian experiment with known standard deviation σ , choose the sample size n such that $n \geq \left(\frac{a\sigma}{\ell}\right)^2$ where $P(Z \leq a) = \frac{1+p}{2}$, $Z \sim G(0, 1)$ to ensure a $100p\%$ confidence interval has width $\leq 2\ell$.
- Understand the differences between a confidence interval for μ where σ is and isn't known. In particular, that it will be wider when we do not assume σ is known.

Hypothesis Testing (Introduction; Section 5.1)

You should understand the fundamentals of hypothesis testing, including:

- Null hypothesis, alternative hypothesis, test statistic or discrepancy measure.
- p -value and its interpretation, including guidelines to be used in STAT 241 for interpreting p -values (see Table 5.1).
- Hypothesis testing for Binomial(n, θ) data, including exact (using the discrepancy measure $D = |Y - n\theta_0|$) and a normal approximation via the Central Limit Theorem.

Hypothesis Testing for Parameters in the $G(\mu, \sigma)$ Model (Section 5.2)

You should be familiar with several specific results relating to the $G(\mu, \sigma)$ model:

- Hypothesis test for $H_0 : \mu = \mu_0$ for the $G(\mu, \sigma)$ model, with unknown standard deviation σ :

$$p\text{-value} = 2 \left[1 - P \left(T \leq \frac{|\bar{y} - \mu_0|}{s/\sqrt{n}} \right) \right]$$

where $T \sim t_{n-1}$ and s = the sample standard deviation.

- Hypothesis test for $H_0 : \sigma^2 = \sigma_0^2$ for the $G(\mu, \sigma)$ model:
 1. Let $u = \frac{(n-1)s^2}{\sigma_0^2}$. Let $U \sim \chi_{n-1}^2$.
 2. If u is large (that is, if $P(U \leq u) > 0.5$) then $p\text{-value} = 2P(U \geq u)$.
 3. If u is small (that is, if $P(U \leq u) < 0.5$) then $p\text{-value} = 2P(U \leq u)$.
- Relationship between confidence intervals and tests of hypothesis. For example, if testing $H_0 : \theta = \theta_0$ results in a $p\text{-value} \leq 0.05$, then a 95% confidence interval for θ will *not* contain θ_0 , and vice-versa.

Likelihood Ratio Tests (Section 5.3)

You should be able to carry out likelihood ratio tests, in particular:

- How the likelihood ratio test relates to the relative likelihood function.
- How to use the likelihood ratio test statistic to test $H_0 : \theta = \theta_0$ for a random variable parameterized by θ .

Gaussian Response Models (Sections 6.1-6.3)

- The simple linear regression model

$$Y_i \sim G(\alpha + \beta x_i, \sigma) \quad i = 1, 2, \dots, n \quad \text{independently}$$

where α, β , and σ are unknown parameters and the x_i are known constants.

- Maximum likelihood estimates and least squares estimates of α and β :

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} \quad \hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

note in particular that you should be able to derive these estimates, but the results themselves would be provided if required.

- The derivation of the distribution of the maximum likelihood estimates of α and β .
- Confidence intervals for β .
- How to test the hypothesis of no linear relationship $H_0 : \beta = 0$.
- Confidence interval for the mean response at x : $\mu(x) = \alpha + \beta x$.
- Prediction interval for new response Y for explanatory variate value x .
- How to check simple linear regression model assumptions using: (1) scatterplot of data and fitted line; (2) residual plot of $(x_i, \hat{r}_i)$; (3) standardized residual plot of $(x_i, \hat{r}_i^*)$; (4) standardized residual plot of $(\hat{\mu}_i, \hat{r}_i^*)$.
- The R^2 statistic (covered in Tutorial 5).

See the summaries in Tables 6.1 and 6.2 in the Course Notes.

Comparing the Means of Two Populations (Section 6.4)

- Point estimators of the means.
- Pooled estimator of variance.
- Confidence interval for the difference between the means assuming equal and unequal variances.
- Confidence interval for the difference between the means in a paired experiment.
- When and why a paired experiment is more appropriate than treating the groups as two independent random samples.

General Gaussian Response Models (Section 6.5)

Note: The majority of this section, as presented in the Course Notes, is not covered in the STAT 231 final. However, you should be familiar with the following.

The general Gaussian response model (also referred to as multiple linear regression) relates explanatory variates $x_{1i}, x_{2i}, \dots, x_{ki}$ to a response Y_i via the model

$$Y_i \sim G(\mu(x_i), \sigma)$$

with $\mu(x_i) = \beta_0 + \sum_{j=1}^k \beta_j x_{ij}$ for $i = 1, 2, \dots, n$, independently.

To test the null hypothesis $H_0 : \beta_j = 0$ we refer to a t -distribution with $n - k - 1$ degrees of freedom.

Multinomial Models (Chapter 7)

- Testing the fit of a model using the Multinomial model and the likelihood ratio or Pearson goodness of fit test statistics (Section 7.1-7.2). In particular you should know how to find the appropriate degrees of freedom for the resulting hypothesis test, and how this is used to find a p -value.
- Testing for independence of variates in two way tables (Section 7.3).

Cause and Effect (Chapter 8)

- Definition of causation.
- Reasons why two variates may be associated, but not necessarily causally related.
- Why randomization can be used to help identify causal relationships.

R Commands

You should be familiar with all R commands introduced in class, in the Introductory R Tutorial, and in R Tutorials 1-4, except for material that is explicitly excluded as discussed in those tutorials. You may be given R commands and the resulting output (numerical and/or graphical) and be asked to interpret it. You may be asked what R command would be appropriate to use to solve a given problem. You may be asked what output would result from running a particular R command.

Exam Front Page Instructions

For your reference, the following information will be provided on the front page of your examination booklet. Note that as this is an open book exam, no formula pages will be provided.

- **Unless a question indicates a numeric value is required**, numeric answers may be left as numeric expressions if desired (e.g., if the correct answer was 2, an answer of ' $1 + 1$ ' would receive the same credit as an answer of '2'). If you express answers as numeric values these should be accurate to at least three decimal places.
- If you are asked to identify the type of a variate, choose from Continuous, Discrete, Categorical, Ordinal, or Complex.
- If you believe insufficient information is provided in a question to determine the requested answer, you may write 'not enough information', 'NEI', or similar. If the question indicates you should justify your answer, this justification is still required even if your answer is 'not enough information'.
- Additional blank pages are provided at the end of the exam for rough work and/or additional space for answers. If you wish any work on these pages to be graded, you must clearly indicate this on the corresponding question page.
- Good luck :)